

## Homework 6 File Submission

- A. Find a formula for  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$  by examining the values of this expression for small values of  $n$ . Then use mathematical induction to prove the formula.

Compute small values of  $n$ :

$$n=1, S(1) = \frac{1}{2}$$

$$n=2, S(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$n=3, S(3) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

Conjecture:

$$S(n) = \sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}$$

Induction Proof:

Base case ( $n=1$ ):  $S(1) = \frac{1}{2} = 1 - \frac{1}{2} = \frac{1}{2}$ . ✓

Inductive hypothesis: Assume for some  $k \geq 1$ ,

$$S(k) = \sum_{i=1}^k \frac{1}{2^i} = 1 - \frac{1}{2^k}$$

Inductive step: We must show  $S(k+1) = \sum_{i=1}^{k+1} \frac{1}{2^i} = 1 - \frac{1}{2^{k+1}}$ .

Starting with the hypothesis,

$$\begin{aligned} S(k+1) &= S(k) + \frac{1}{2^{k+1}} = \left(1 - \frac{1}{2^k}\right) + \frac{1}{2^{k+1}} \\ &= 1 - \frac{2}{2^{k+1}} + \frac{1}{2^{k+1}} = 1 - \frac{1}{2^{k+1}} \end{aligned}$$

Conclusion:  $\therefore$  Since the base case holds and  $S(k) \rightarrow S(k+1)$  holds,

by mathematical induction,  $\sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}$  for all  $n \in \mathbb{Z}^+$ . □

- B. Prove that  $3^n < n!$  if  $n$  is an integer greater than 6.

Induction Proof:

Base case ( $n=7$ ):  $3^7 = 2187$ ,  $7! = 5040$ . ✓

Since  $3^7 < 7!$ , the base case holds.

Inductive hypothesis: Assume for some  $k \geq 7$  that  $3^k < k!$ .

Inductive step: We must show  $3^{k+1} < (k+1)!$ .

Starting from the hypothesis,  $3^{k+1} = 3 \cdot 3^k < 3 \cdot k!$ .

Since  $k \geq 7$ , we have  $k+1 \geq 8 > 3$ , then

$$3 \cdot k! \leq (k+1) \cdot k! = (k+1)!$$

Therefore,  $3^{k+1} < (k+1)!$ .

Conclusion:  $\therefore$  Since the base case holds and  $3^k < k! \rightarrow 3^{k+1} < (k+1)!$  holds,

by mathematical induction,  $3^n < n!$  for all integers  $n > 6$ . □